

NLP-Based Customer Support Ticket Classification System

Milestone 3: Model Evaluation

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1. Performance Metrics

For our classification system project we chose Accuracy, F1-score, Precision and Recall as the metrics to use for model evaluation. We also reviewed the Confusion Matrix to get better insight into the results.

- **Accuracy** measures the overall percentage of correctly classified support tickets.
- **Precision** measures how often the predicted ticket category is correct.
- **Recall** measures the model's ability to identify all instances of each ticket category.
- **F1-score** provides a balanced measure by combining precision and recall into a single metric.

Two model variants were trained for evaluation: Naïve Bayes with TF-IDF vectorization and Naïve Bayes with Bag of Words vectorization. We used these results to compare the two models for final model selection.

SUMMARY OF RESULTS:

Model	Feature Representation	Model Stage	Accuracy	Precision	Recall	F1 Score
Naive Bayes	TF-IDF	Training	0.4902	0.4946	0.4902	0.4891
Naive Bayes	Bag-of-Words	Training	0.5635	0.5648	0.5635	0.5637
Naive Bayes	TF-IDF	Testing	0.1742	0.1720	0.1742	0.1713
Naive Bayes	Bag-of-Words	Testing	0.1882	0.1869	0.1882	0.1870

Additional details can be seen in the following pages.

a) Training Evaluation: The model training evaluation metrics for both are shown below.

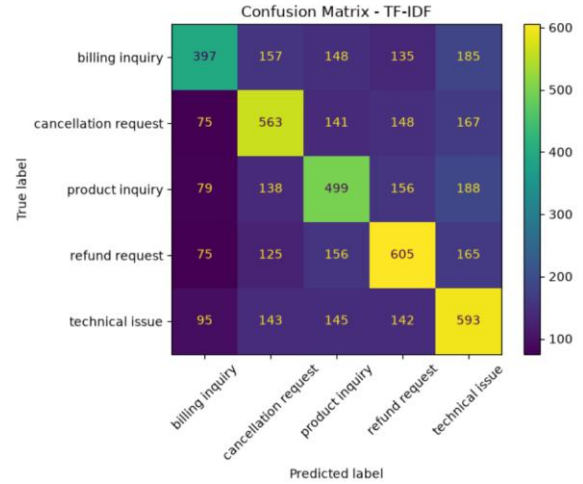
Naïve Bayes with TF-IDF Vectorization

TF-IDF Naive Bayes Evaluation

Accuracy : 0.4902
Precision: 0.4946
Recall : 0.4902
F1 Score : 0.4891

Classification Report:

	precision	recall	f1-score	support
billing inquiry	0.55	0.39	0.46	1022
cancellation request	0.50	0.51	0.51	1094
product inquiry	0.46	0.47	0.46	1060
refund request	0.51	0.54	0.52	1126
technical issue	0.46	0.53	0.49	1118
accuracy			0.49	5420
macro avg	0.50	0.49	0.49	5420
weighted avg	0.49	0.49	0.49	5420



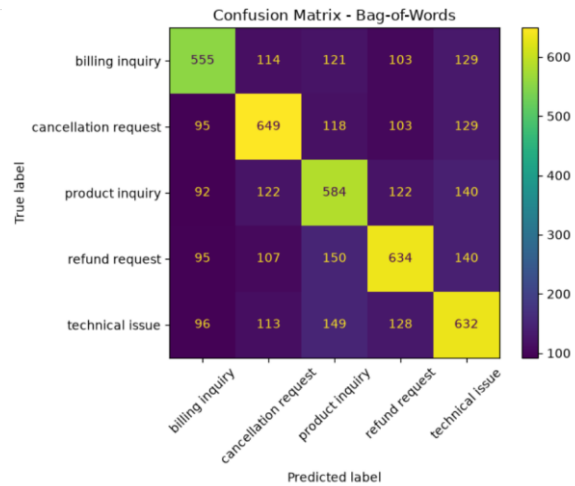
Naïve Bayes with Bag of Words Vectorization

Bag of Words Naive Bayes Evaluation

Accuracy : 0.5635
Precision: 0.5648
Recall : 0.5635
F1 Score : 0.5637

Classification Report

	precision	recall	f1-score	support
billing inquiry	0.59	0.54	0.57	1022
cancellation request	0.59	0.59	0.59	1094
product inquiry	0.52	0.55	0.54	1060
refund request	0.58	0.56	0.57	1126
technical issue	0.54	0.57	0.55	1118
accuracy			0.56	5420
macro avg	0.56	0.56	0.56	5420
weighted avg	0.56	0.56	0.56	5420



b) Test Evaluation: The model test evaluation metrics for both are shown below.

Naïve Bayes with TF-IDF Vectorization

TF-IDF Naive Bayes Evaluation

Accuracy : 0.1742

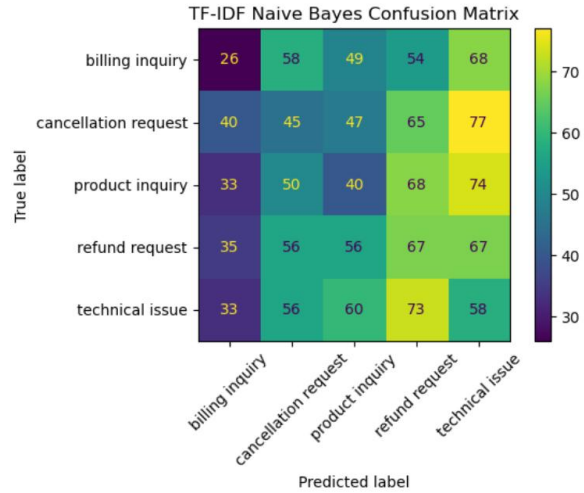
Precision: 0.1720

Recall : 0.1742

F1 Score : 0.1713

Classification Report

	precision	recall	f1-score	support
billing inquiry	0.16	0.10	0.12	255
cancellation request	0.17	0.16	0.17	274
product inquiry	0.16	0.15	0.15	265
refund request	0.20	0.24	0.22	281
technical issue	0.17	0.21	0.19	280
accuracy			0.17	1355
macro avg	0.17	0.17	0.17	1355
weighted avg	0.17	0.17	0.17	1355



Naïve Bayes with Bag of Words Vectorization

Bag-of-Words Naive Bayes Evaluation

Accuracy : 0.1882

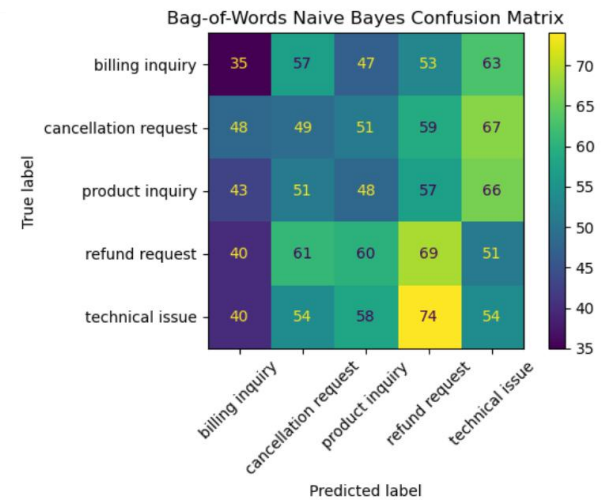
Precision: 0.1869

Recall : 0.1882

F1 Score : 0.1870

Classification Report

	precision	recall	f1-score	support
billing inquiry	0.17	0.14	0.15	255
cancellation request	0.18	0.18	0.18	274
product inquiry	0.18	0.18	0.18	265
refund request	0.22	0.25	0.23	281
technical issue	0.18	0.19	0.19	280
accuracy			0.19	1355
macro avg	0.19	0.19	0.19	1355
weighted avg	0.19	0.19	0.19	1355



2. Analysis and Insights:

a) Interpretation of the Results:

- **TRAINING RESULTS:**

Both models perform well on the training data with an accuracy of 0.49 and 0.56 respectively.

Given that the dataset contains five relatively balanced categories, the performance is better than the 20% accuracy expected from random guessing. This indicates that the model is able to identify meaningful relationships between the vectorized features and the label categories.

- **TEST RESULTS:**

Both models perform poorly on the test data, having an accuracy of 0.17 and 0.19 respectively.

A low score for all four metrics (accuracy, precision, recall, and F1 Score) indicates that the model is performing poorly across the board. Because the metrics are all similarly low, the model is not effectively distinguishing between the different ticket categories. The accuracy of 0.17 – 0.19 is close to the random baseline of 0.20 for a five-class problem. This suggests that the model is learning little more than chance-level predictions.

The consistently low precision, recall, and F1 scores across classes indicate that the model struggles both to correctly identify each category and to make reliable predictions for those categories.

- **FINAL VECTORIZATION SELECTION:**

The results indicate that the Bag-of-Words feature representation was more effective than TF-IDF for this dataset when used with the Multinomial Naive Bayes classifier. Although the performance improvement was small, it consistently produced higher Accuracy, Precision, Recall, and F1-score. This model was chosen for use with future support ticket classification.

b) Model Strengths:

The Naïve Bayes model is simple and efficient. It trained quickly and provided a baseline for automatically classifying customer support tickets.

The model shows consistency in the ability to predict across all 5 categories, with balanced performance, within the training data. This can be seen in the evaluation

metrics above with accuracy, precision, recall and the F1 score being similar for each category. This suggests that the model has learned patterns that are useful for distinguishing among the categories, even if its overall ability to accurately classify individual tickets remains fairly low (as seen in the test data).

c) **Model Limitations:**

On the whole, there is a large gap between the training and testing results evaluation, which suggests there could be overfitting and indicates that further improvements are needed to improve generalization to unseen customer support tickets.

The overall low accuracy indicates that the model had difficulty distinguishing between similar ticket categories, suggesting that additional training data or more advanced models may improve performance.

d) **Suggested Improvements:**

Future work could include hyperparameter tuning to optimize the Naive Bayes classifier, experimenting with n-grams and additional preprocessing techniques to better capture contextual information, balancing the dataset if needed, and evaluating more advanced classification models such as Support Vector Machines, Logistic Regression, or transformer-based models like BERT to improve classification performance. These approaches may improve the model's ability to generalize to unseen customer support tickets and achieve higher classification accuracy.

e) **Ablation Studies**

An ablation study was not conducted as part of this project due to time constraints. However, future work could systematically evaluate the contribution of each preprocessing and the feature extraction step. For example, separate experiments could compare model performance without stop-word removal, lemmatization, punctuation removal, and lowercasing to determine which preprocessing techniques have the greatest impact on classification accuracy.

3. Error Analysis:

- **Common Mistakes:**

When looking at the data from the confusion matrices for the test data, we can see that there is a fairly even distribution of incorrect predictions for each of the categories. This suggests that the model cannot successfully distinguish between the categories. This could be due to similarity in some of the vocabulary used in the customer complaints across categories, or the varying lengths of the complaint text. Some are simple, and others contain various details that could have it fit into multiple categories.

- **Failure Case Analysis:**

When reviewing some of the records, there appear to be disconnects between the vocabulary in the complaint and the category assigned.

1st example:

Complaint Field: “i've noticed a software bug in the fitbit charge app. it's causing data loss and unexpected errors. how can i resolve this issue? i need assistance as soon as possible because it's affecting my work and productivity. resolution pending”

Category: billing inquiry

2nd example:

Complaint Field: “i'm having an issue with the nest thermostat. please assist. please. please. i need assistance as soon as possible because it's affecting my work and productivity. more left involve agree station billion would.”

Category: refund request

In both of these examples, the complaint vocabulary aligns more with a technical issue than a billing inquiry or refund request.

If data points like these are used in the training set, the model could incorrectly select vocabulary, better suited for another category, causing issues with overlapping requirements. If data points like this are used in a test set where the model was properly fit using billing or refund related vocabulary, these would likely be mis-classified.

- **Strategies for Improvement:**

The observed error patterns suggest several opportunities for improvement. Additional preprocessing techniques, such as incorporating bigrams or trigrams, could help preserve important phrases instead of treating words independently. Hyperparameter tuning and feature selection may also improve the classifier's ability to distinguish between similar ticket categories. Finally, evaluating more advanced classification models, such as Logistic Regression, Support Vector Machines (SVMs), BiLSTM, or transformer-based models like BERT with pre-trained text, may improve the model's ability to capture contextual relationships within customer support tickets and reduce misclassification rates.